



Weekly Updates 4/4

Team 3: Robotic Assistant for MRI-Guided Low Back Pain Injections

High Level Updates

- Website updated
- Robot Working!!
 - Addressed friction on Motor F screw and Motor A encoder faults
- Project Update Presentation 4/17
 - [Outline done](#)



Tyler: Updates



Tyler: Problems encountered & Next Steps

Problems

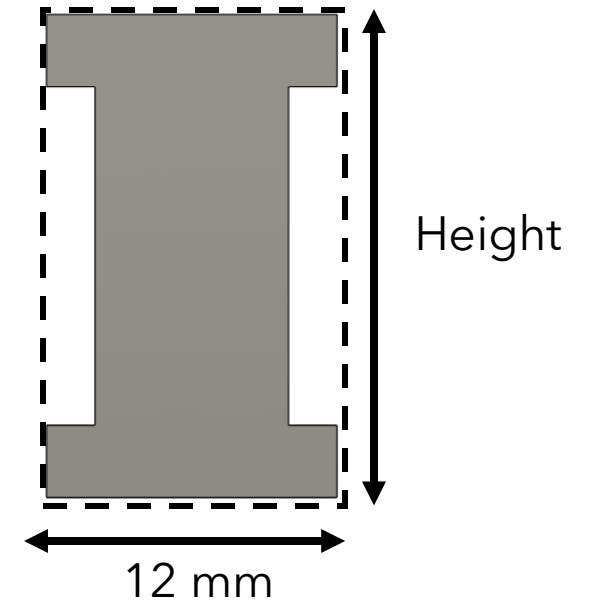
Next Steps



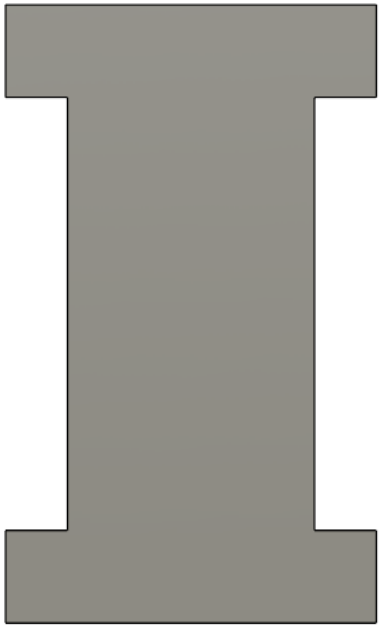
Tyler: Deliverables by Next Week

Sai: Updates

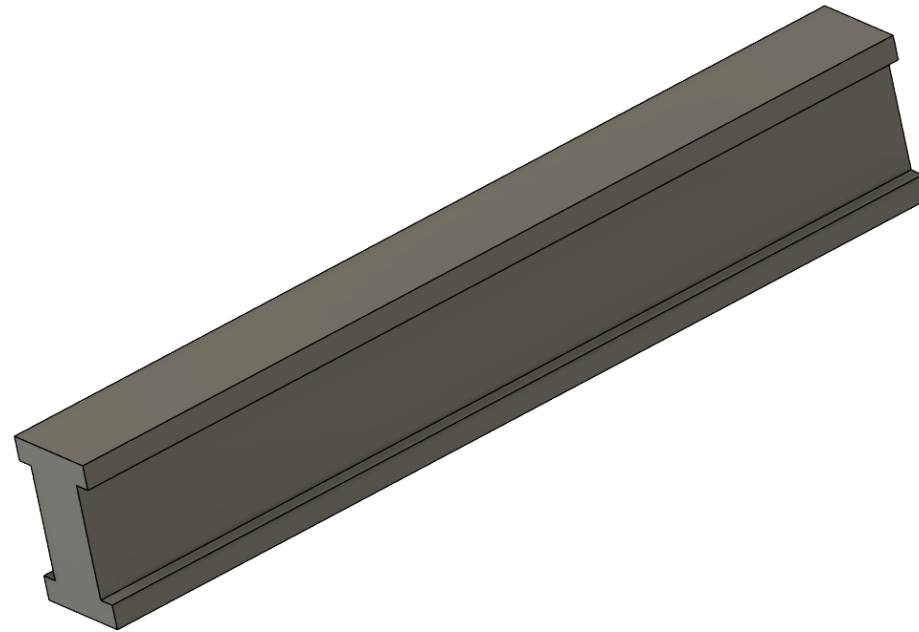
- Changing the profiles of the linkage to get more stiffness.
- Constraints:
 - Material: ABS
 - Bolt Hole Dimensions to not be changed.
 - Length (111 mm) and Width (12 mm) of the linkage remains unchanged.



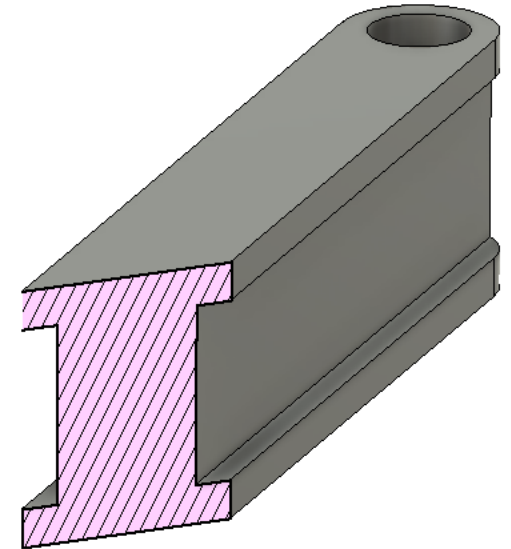
Profile 1: I Beam Section



Profile View

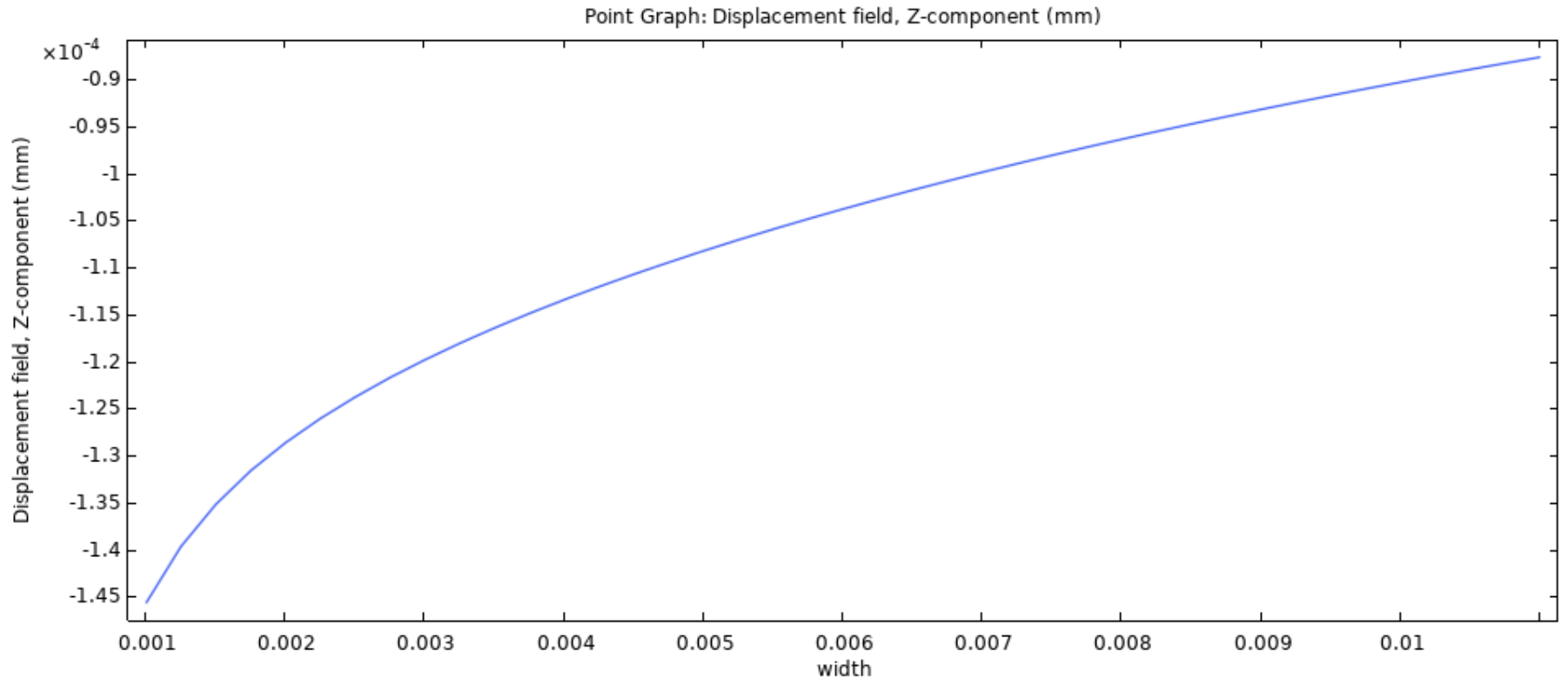
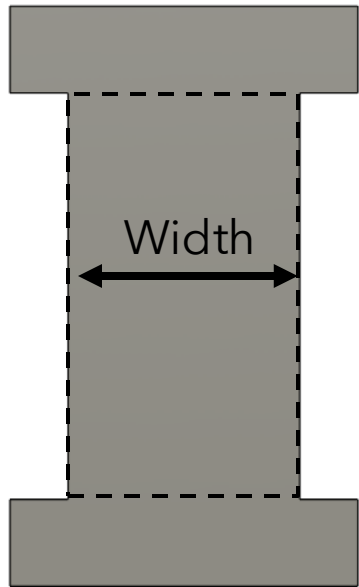


Profile Side
View



Linkage View

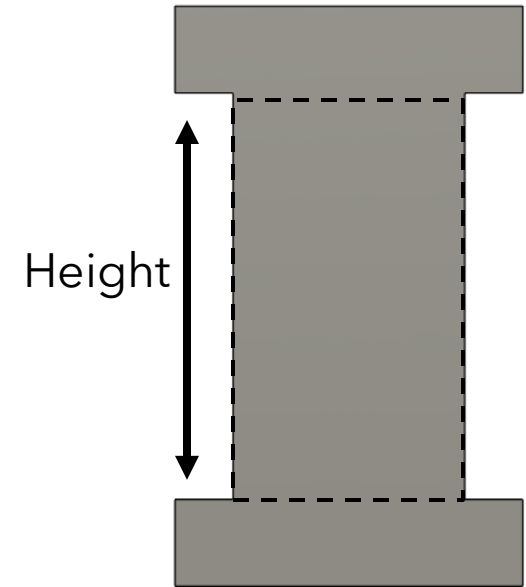
Attempt 2



- As expected, as the width increases the rigidity of the linkage increases

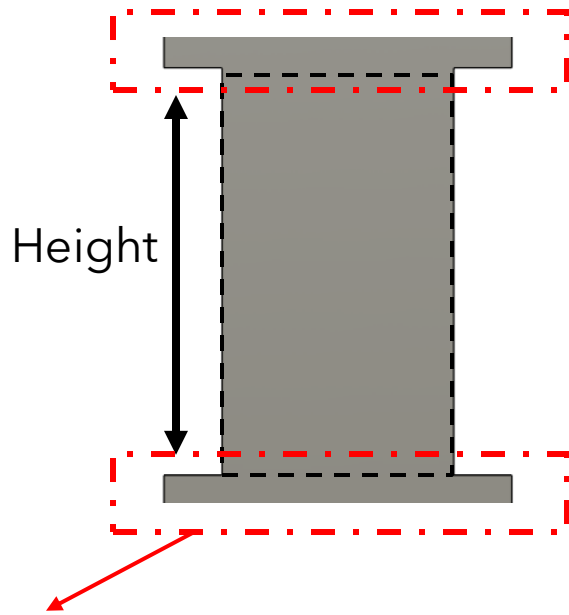
Attempt 3

- Utilizing the results of the previous iteration, we choose a suitable width such that the I beam profile retains its contour
- Hence, we choose the width to be 7 mm
- In this attempt, we are trying to see what is the required height of the linkage to match the deflection of the 5 mm and 6 mm Carbon Fiber Linkages

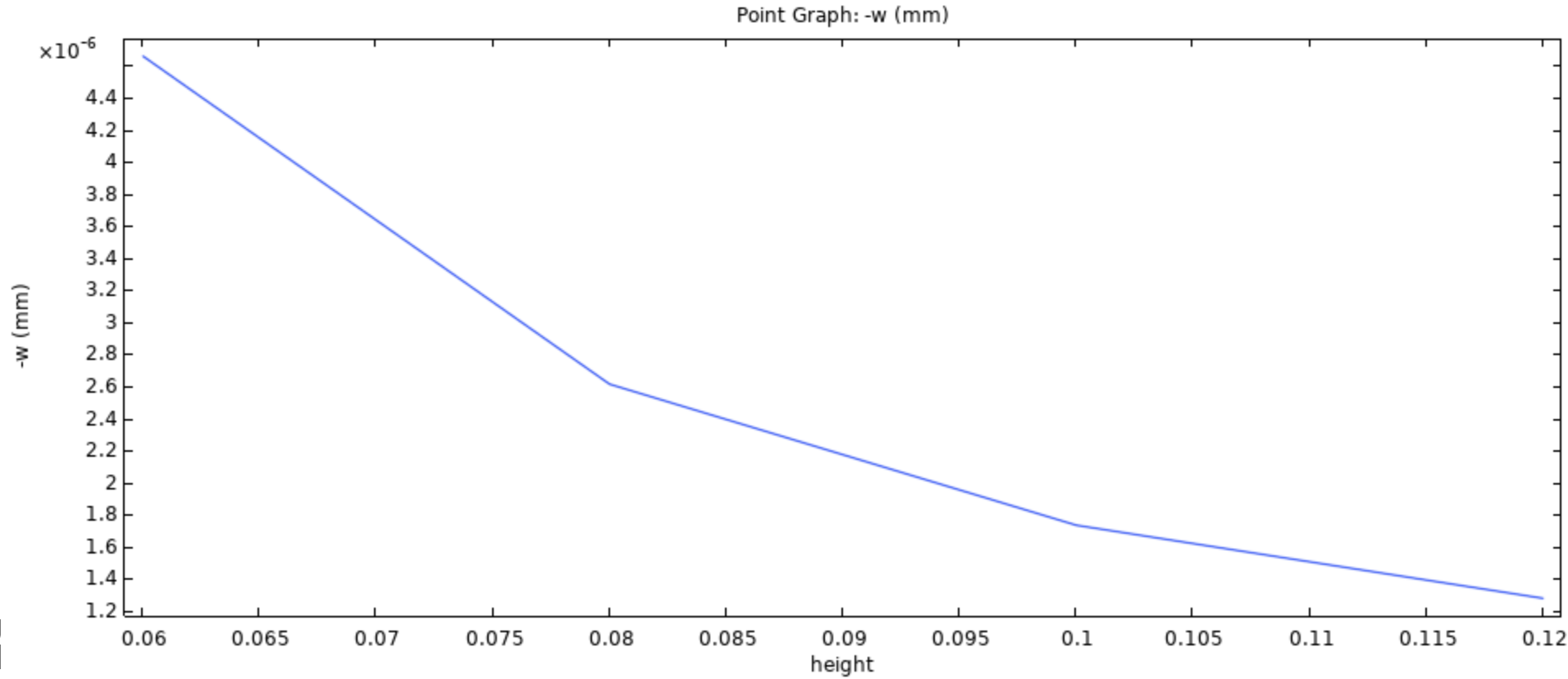


Attempt 3

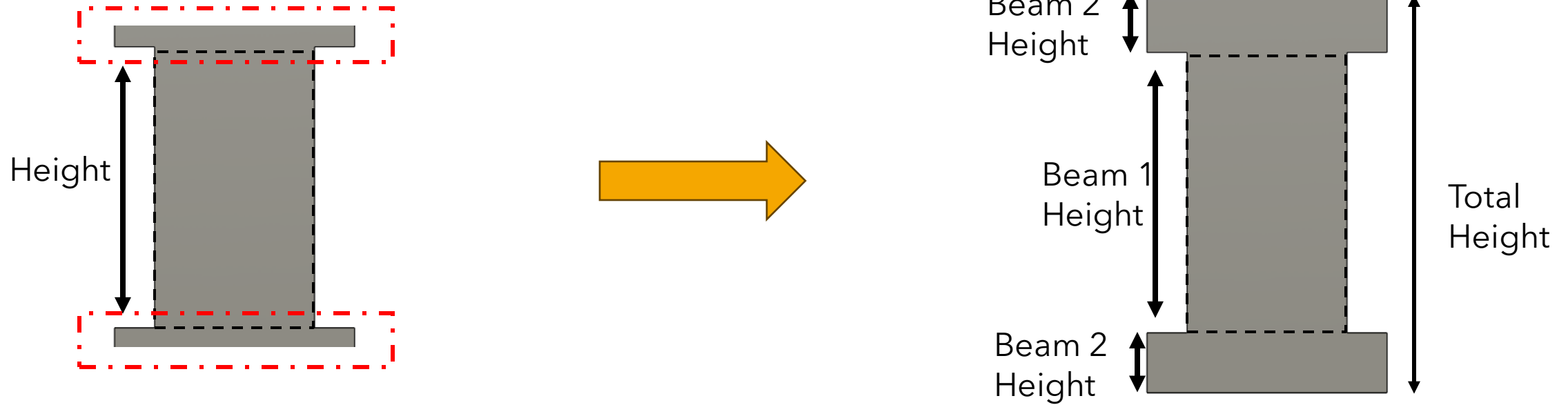
- Problematic Profiles



As these sections were not being scaled, it wouldn't be considered true representation.

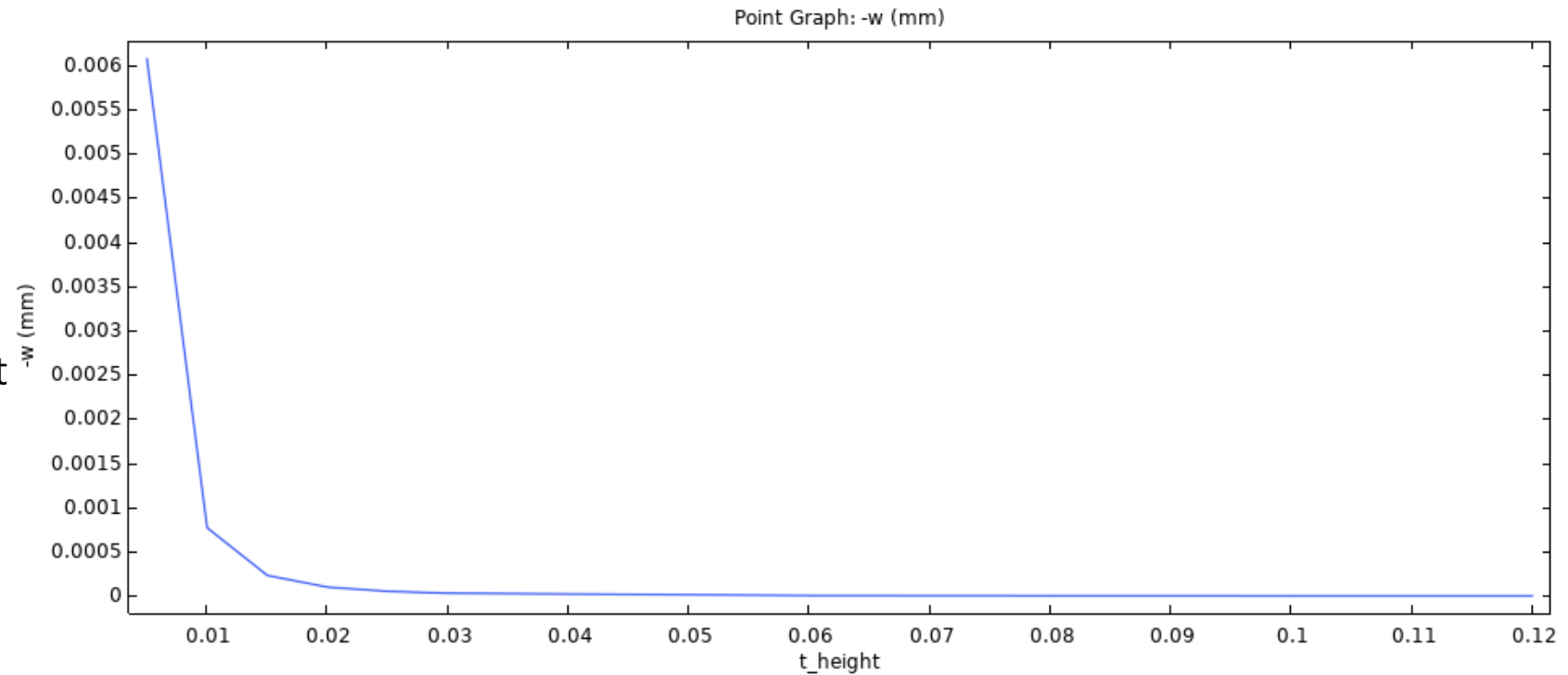
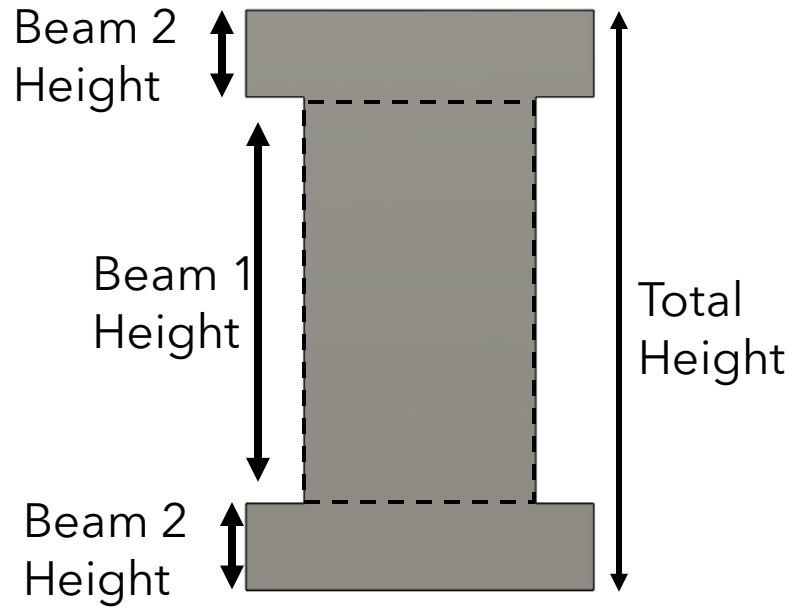


Attempt 4



- Beam 1 Height: 70% of Total Height
- Beam 2 Height: 15% of Total Height

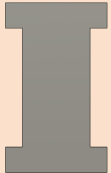
Attempt 4



Results Comparison

Material	Deflection(10 ⁻⁶ m)
Carbon Fiber	
5 mm	9.7
6 mm	8.1
ABS	
Attempt 3 (100 mm)	1.7
Attempt 4 (100 mm)	1.6

Attempt 1

Profile	Second Moment of Inertia	Weight (kg)	Ratio(Inertia:Weight)	Deflection under 5 N (mm)
	1.55E-8	1.82E-5	8.52E-4	1.1E-4
	1.68E-8	1.76E-5	9.54E-4	1.12E-4
	1.62E-8	1.7E-5	9.53E-4	1.13E-4
	1.40E-8	1.38E-5	10.01E-4	1.3E-4
	1.59E-8	1.62E-5	9.9E-4	1.2E-4
	1.64E-8	1.73E-5	9.5E-4	1.13E-4

Diana: Updates

- Robot is fully working!!
- Robot Operation Procedure Draft
- Optical Tracker Mounts Done and ready for Analysis
- Lower Body Mount Progress

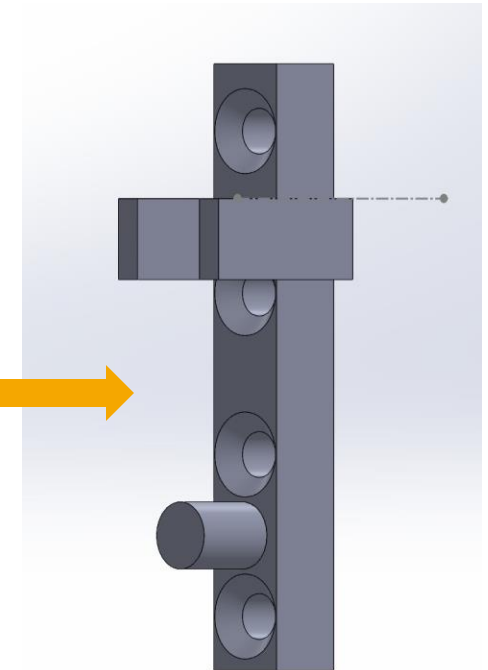
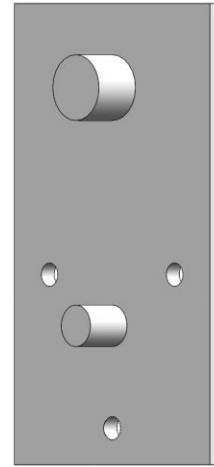
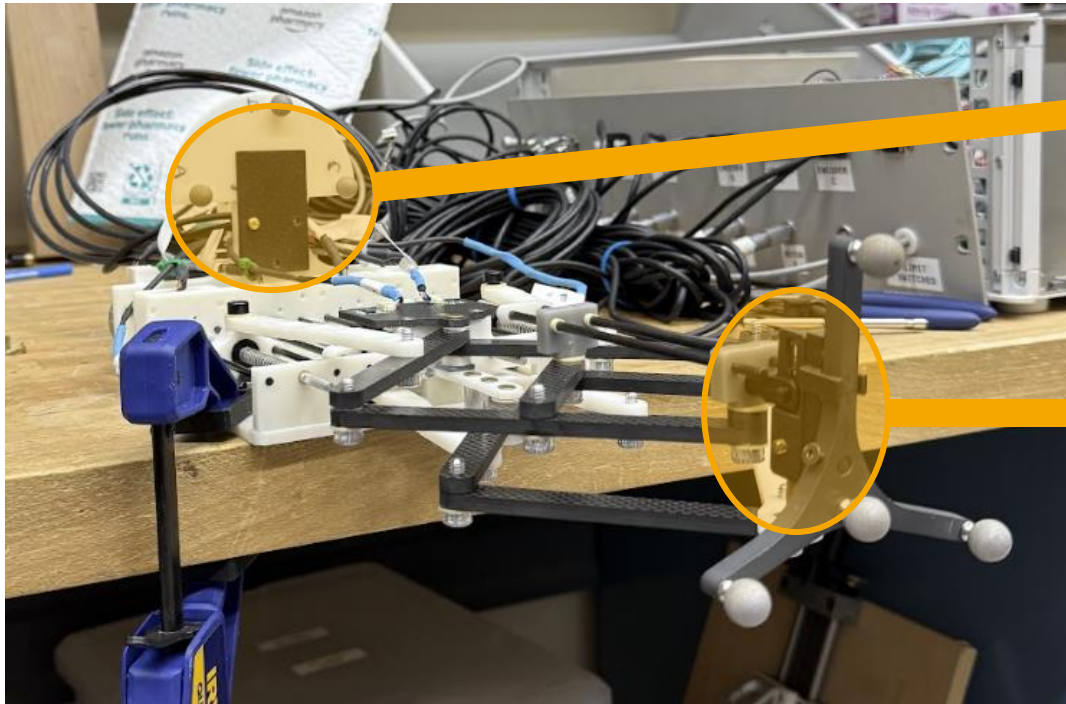
Diana: Robot Working

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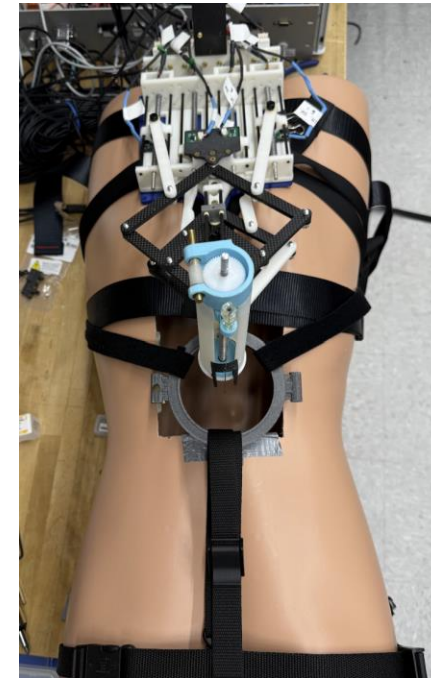
Diana: Robot Operation Procedure

- Aabhas and I developed a means to ensure the robot will be working
- Roughly operates as:
 - Digital I/O all motors back, ensure encoders working
 - Home 3x
 - Move 1mm back
 - Move to Motor Max and Min (-5mm)

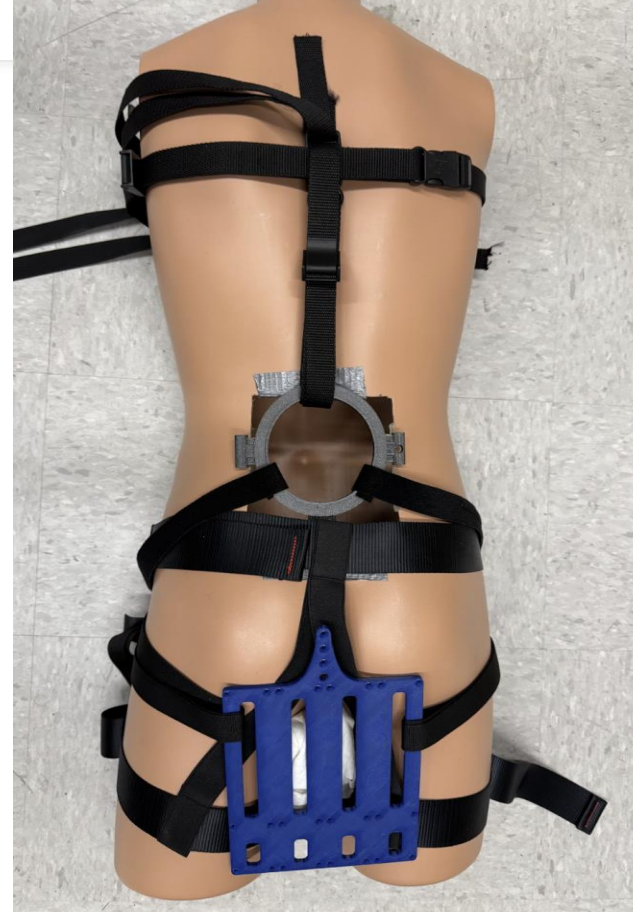
Diana: Optical Tracker Mounts



Diana: Lower Body Mount



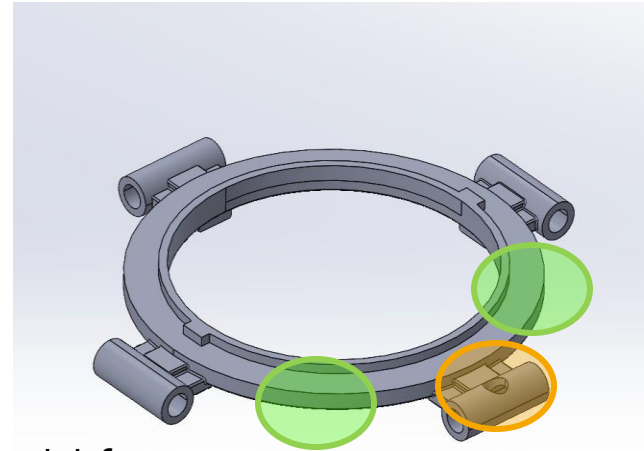
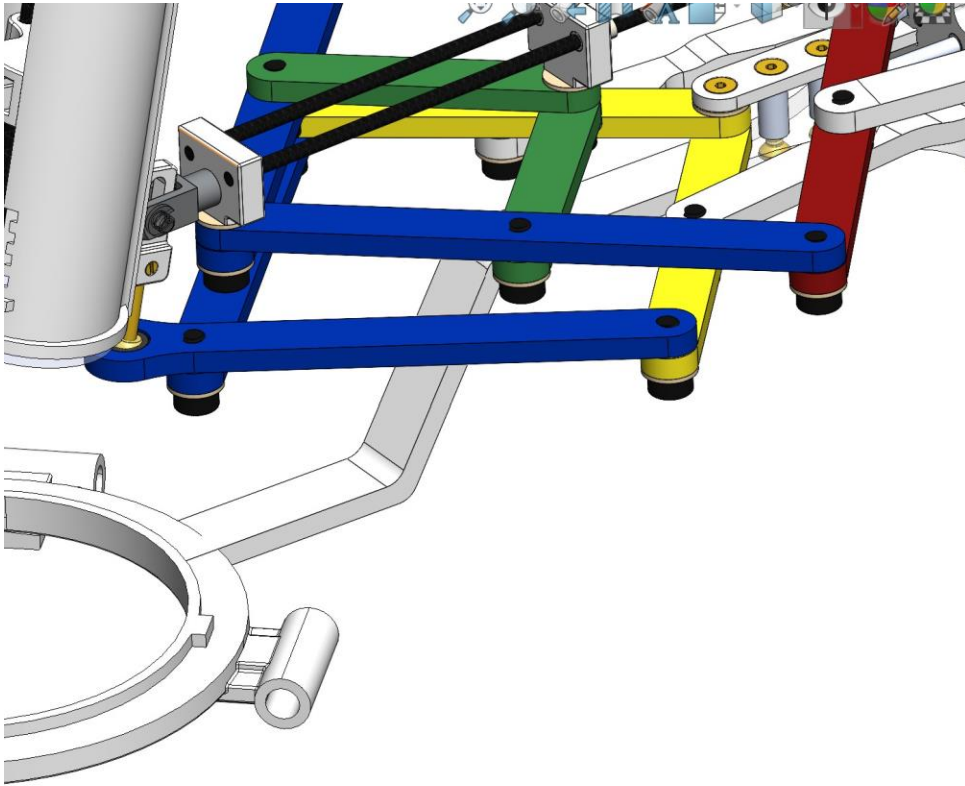
Diana: Lower Body Mount



Diana: Lower Body Mount ***NEXT STEPS***



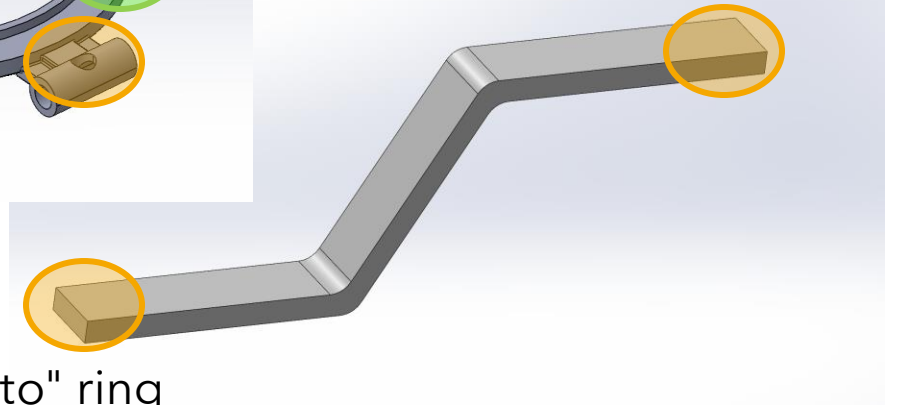
Diana: Lower Body Mount ***NEXT STEPS***



Add fixture to
hold strap/velcro

Add tab to "plug-into" ring

Add hole to screw
into robot



Diana: Problems encountered & Next Steps

Problems

- Ghost in the lab
 - Encoders working then not, then working again

Next Steps

By 4/19

- Update Procedures/Documentation
- Assist with End Position Optical Tracking
- Updated CAD for Lower Body Mount Parts
- Finalized General Lower Body Mount

Full Team Next Steps

- Checkpoint 4/17
- Optical Tracking Testing – Endpoint Positioning Accuracy
- Website Documentation up to date
 - All procedures
 - User documentation
 - Milestone update